





Lecture 11 - The Chain Rule

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10/15/2026

Quiz Time

Let’s get into neat rows, spread out, and slay this quiz! :)

The Chain Rule

The Road So Far

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- Power Rule: $\frac{d}{dx} x^n$
- Trigonometric Rules: $\frac{d}{dx} \sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx} e^x$, $\frac{d}{dx} \ln(x)$ ⇝ Tuesday

Combination Rules:

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Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

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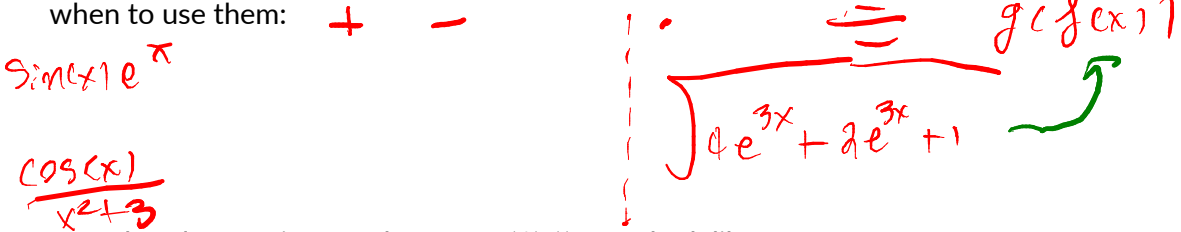
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But what do functions of the form $g(f(x))$ even look like?

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- x^n power
- c^x exponential
- $\sin(x), \cos(x)$, etc trig
- $|x|$ abs
- $\log_c(x)$ log.

$e^{9x} + e^{4x} + 1$

$2 \cdot 4 = 8$

$(e^{4x})^2 + (e^{4x})^1 + 1 \cdot (e^{4x})^0$

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What is the Chain Rule? II

$u = 4x$ $\quad e^u$ $\quad u = e^{4x}$

$h(x) = 4x$ $\quad f(x) = e^x$ $\quad g(f(h(x)))$

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When we *compose* functions (do the $g(f(x))$ thing), what does this actually do?
We make one function's output another function's input instead of just inputting x!

$u^2 + 2u + 1$

$e^{8x} + 2e^{4x} + 1 = g(f(x))$

$(e^{4x})^2$ $\quad 1 = (e^{4x})^0$

$g(x) = e^{4x}$

$g(x) = x^2 + 2x + 1$

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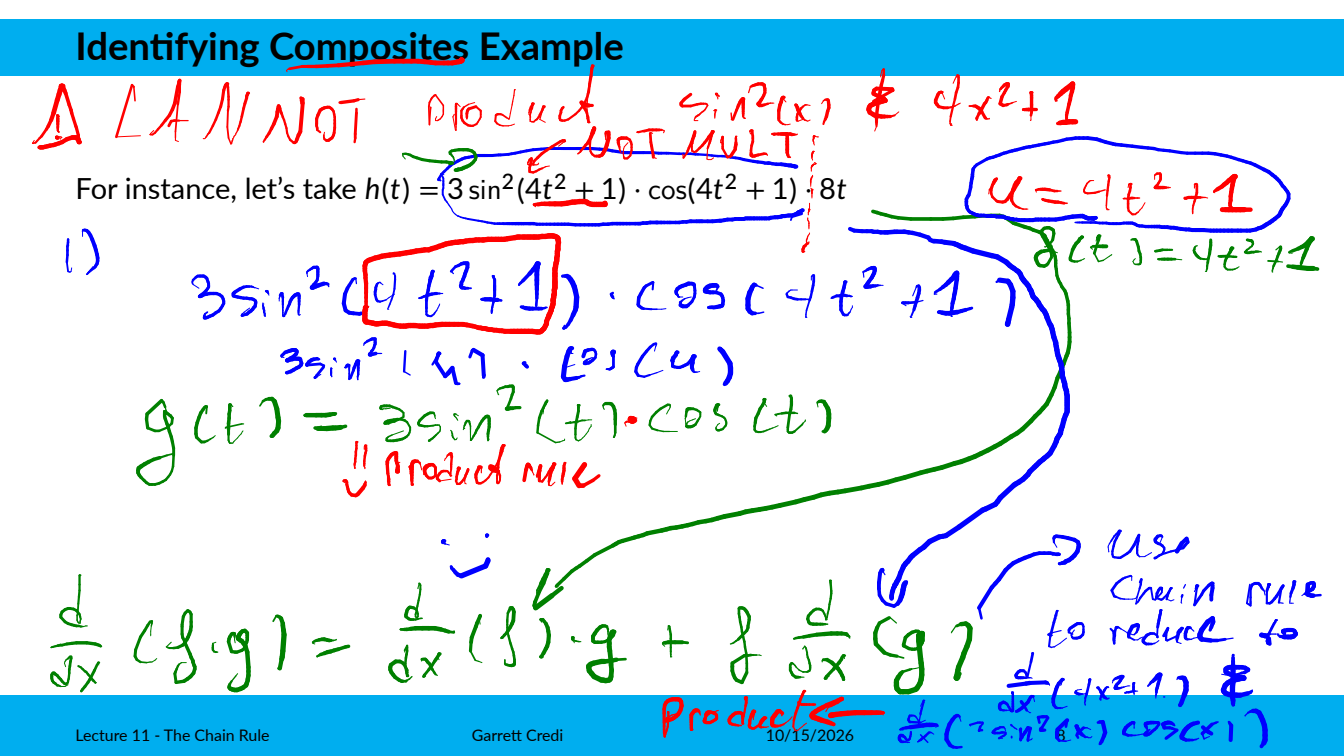
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2. In each of the smaller pieces, identify where the variables actually are.
3. At each occurrence of a variable, find the biggest/longest expression that it lives in - this is the innermost function.
4. Treat that whole expression as a new variable, and repeat from step 1.

Identifying Composites Example



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We start by linearizing *only* $f(x)$,

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

$$g(f(x))$$

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We know that $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$, so we end up wanting to further linearize:

$g(f(x)) \approx g(f(x_0) + f'(x_0)(x - x_0))$

how can I linearize g around some point

Deriving Chain Rule II

Note: when we linearize $f(x)$ near $x = x_0$, we mean that for values $x \approx x_0$ we'll know that $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$

Deriving Chain Rule II

$$g(f(x_0) + f'(x_0)(x - x_0))$$

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When $x \approx x_0$, what is $f(x_0) + f'(x_0)(x - x_0)$ close to?

≈ 0

$$f(x_0) + f'(x_0)(x - x_0) \approx f(x_0)$$

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When $x \approx x_0$, what is $f(x_0) + f'(x_0)(x - x_0)$ close to?

Since $f(x_0) + f'(x_0)(x - x_0) \approx f(x_0)$, we will linearize $g(x)$ around $f(x_0)$ instead!

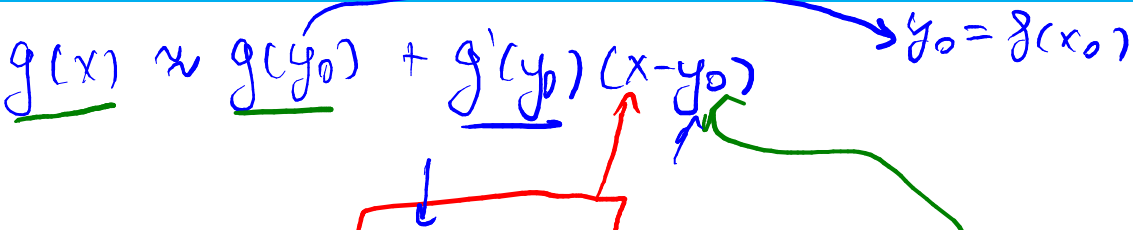
$g(f(x_0) + f'(x_0)(x - x_0)) \approx g(f(x_0))$

Deriving Chain Rule III

$g(x) \approx g(y_0) + g'(y_0)(x - y_0)$ $\quad y_0 = g(x_0)$

$$g(f(x)) \approx g(f(x_0) + f'(x_0)(x - x_0)) \quad \text{linearize } g \text{ around } f(x_0) \tag{2}$$

Deriving Chain Rule III



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Deriving Chain Rule III

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$$\approx g(f(x_0)) + g'(f(x_0))(f(x_0) + f'(x_0)(x - x_0)) - f(x_0)) \tag{3}$$

$$= g(f(x_0)) + g'(f(x_0))(f'(x_0)(x - x_0) + f(x_0) - f(x_0)) \tag{4}$$

Deriving Chain Rule III

$g(x) \approx g(y_0) + g'(y_0)(x - y_0)$ $\quad y_0 = f(x_0)$

$$
\begin{aligned}
g(f(x)) &\approx g(f(x_0) + f'(x_0)(x - x_0)) && \text{linearize } g \text{ around } f(x_0) && (2)\\
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&= g(f(x_0)) + g'(f(x_0))f'(x_0)(x - x_0) && && (5)
\end{aligned}
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Deriving Chain Rule IV

The derivative is the slope of the best linear approximation $\Longrightarrow$

The Chain Rule

The Chain Rule (Leibniz's Version)

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$$\frac{d}{dx}g(f(x)) = g'(f(x))f'(x) \tag{6}$$

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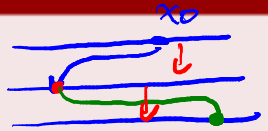
The Chain Rule

$$\frac{d}{dx}g(f(x)) = g'(f(x))f'(x) \tag{6}$$

Or, written differently

The Chain Rule (Leibniz's Version)

$$\frac{d(g \circ f)}{dx}\Big|_{x=x_0} = \frac{dg}{dx}\Big|_{x=f(x_0)} \cdot \frac{df}{dx}\Big|_{x=x_0} \tag{7}$$



Chain Rule Example I

$f(t) = 4t^2 + 1 \quad g(t) = 3\sin^2(t)\cos(t)$

Let's compute the derivative of that function from before:

$$h(t) = 3\sin^2(4t^2 + 1) \cdot \cos(4t^2 + 1) \cdot 8t$$

$g'(f(t))f'(t)$ (8)

$8 \cdot (3\sin^2(4t^2+1) \cdot \cos(4t^2+1)) +$

$8t($ $8t$ Exercise $)$

Chain Rule Example II

We can also derive the quotient rule in terms of the chain rule since

$$\frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)} = f(x) \cdot \left(\frac{1}{x} \circ g\right)(x) \tag{9}$$

i.e. $\frac{1}{g(x)}$ is a composite function! You first do $g(x)$ and then do the function $\frac{1}{x}$.

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$$= \frac{f'(x)}{g(x)} + f(x) \cdot \frac{-1}{g(x)^2} \cdot g'(x) = \frac{f'(x)g(x)}{g(x)^2} - \frac{f(x)g'(x)}{g(x)^2} \tag{12}$$

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$$= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} \tag{13}$$

Chain Rule Example III

We can also frame this more in terms of real world applications, too long for a slide but it's in the worksheet.

Worksheet Time

Let’s get into groups and work on the worksheet!

End of Class

That’s all folks!